



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE:24AD3107 DATA SCIENCE

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TITLE OF THE PROJECT	EVision – Electric Vehicle Charging Demand Analytics	
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Project Title: “EVision – Electric Vehicle Charging Demand Analytics”

Abstract:

The rapid adoption of electric vehicles (EVs) is increasing the demand for efficient and reliable charging infrastructure. As the number of electric vehicles grows, charging stations may experience uneven usage, resulting in peak-time congestion, underutilized charging facilities, and difficulty in managing energy demand. Understanding charging patterns and predicting future demand can help charging station operators plan resources more effectively and improve the overall charging experience for EV users.

The proposed **EVision – Electric Vehicle Charging Demand Analytics** system is a data-driven analytics platform designed to analyze historical electric vehicle charging data and identify patterns in charging behavior. The system will collect and preprocess charging-related data such as charging date and time, charging duration, energy consumed, charging station, number of charging sessions, and other relevant attributes. Data cleaning techniques will be applied to handle missing values, duplicate records, inconsistent data, and other data quality issues before performing exploratory data analysis.

The system will perform comprehensive exploratory and statistical analysis to identify peak charging hours, high-demand periods, station utilization patterns, average charging duration, and energy consumption trends. Visualization techniques will be used to present daily, weekly, and monthly charging patterns through interactive charts and dashboards. Correlation analysis will also be performed to understand relationships between charging duration, energy consumption, session frequency, and other relevant variables.

Machine learning and predictive analytics techniques will then be applied to estimate future charging demand. Suitable regression or machine learning models will be trained using historical charging data and evaluated using appropriate performance metrics. The system will analyze the model results and provide predictions regarding expected charging demand during different time periods. This can help identify periods of high charging demand and support better planning of charging infrastructure and energy resources.

An interactive dashboard will be developed using Python-based visualization and dashboard technologies such as **Pandas, Matplotlib/Seaborn, Plotly, Scikit-learn, and Streamlit**. The dashboard will provide key indicators, charging trends, station-wise analysis, peak-demand periods, energy consumption patterns, and predicted charging demand in an easy-to-understand format.

Overall, EVision demonstrates the practical application of data cleaning, exploratory data analysis, statistical analysis, data visualization, machine learning, and predictive analytics to address a real-world electric vehicle infrastructure problem. The proposed system can provide useful data-driven insights to charging station operators and infrastructure planners for understanding demand patterns, managing peak periods, and supporting the efficient utilization of EV charging resources.